Retail Sales Performance Analysis & Profitability Insights Report

Business-Oriented Exploratory Data Analysis

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Executive Summary

This project presents a structured end-to-end analysis of a retail sales dataset to evaluate revenue trends, profitability drivers, and operational risks.

Key Findings:

• Revenue shows steady year-over-year growth
• Technology is the most profitable category
• High discount levels significantly reduce profit margins
• A small group of sub-categories drives a large portion of revenue
• A measurable number of transactions generate negative profit

Strategic Recommendations:

• Optimize discount policies to protect margins
• Prioritize high-profit categories and sub-categories
• Reduce loss-making transactions
• Implement region-specific growth strategies

This notebook demonstrates a complete workflow from raw data cleaning to business insight generation.

1. Business Objective

The purpose of this analysis is to:

- Assess overall sales performance
- Identify profitability drivers
- Evaluate the impact of discounting
- Analyze category and regional contribution
- Generate actionable strategic insights

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', None)
sns.set_style("whitegrid")
```

In [2]:
```
df = pd.read_csv('/kaggle/input/datasets/bravehart101/sample-supermarket-datas
df.head()
```

Out[2]:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category
0	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
1	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
2	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies
3	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture
4	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies

In [3]:
```
df.tail()
```

Out[3]:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Categ
9989	Second Class	Consumer	United States	Miami	Florida	33180	South	Furni
9990	Standard Class	Consumer	United States	Costa Mesa	California	92627	West	Furni
9991	Standard Class	Consumer	United States	Costa Mesa	California	92627	West	Techno
9992	Standard Class	Consumer	United States	Costa Mesa	California	92627	West	O Sup
9993	Second Class	Consumer	United States	Westminster	California	92683	West	O Sup

2. Data Understanding

In [4]:
```
df.shape
```

Out[4]: (9994, 13)

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 13 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   Ship Mode     9994 non-null   object 
 1   Segment       9994 non-null   object 
 2   Country       9994 non-null   object 
 3   City          9994 non-null   object 
 4   State         9994 non-null   object 
 5   Postal Code   9994 non-null   int64  
 6   Region        9994 non-null   object 
 7   Category      9994 non-null   object 
 8   Sub-Category  9994 non-null   object 
 9   Sales         9994 non-null   float64
 10  Quantity      9994 non-null   int64  
 11  Discount      9994 non-null   float64
 12  Profit        9994 non-null   float64
dtypes: float64(3), int64(2), object(8)
memory usage: 1015.1+ KB
```

In [6]:
```
df.describe()
```

Out[6]:

	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	55190.379428	229.858001	3.789574	0.156203	28.656896
std	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	90008.000000	209.940000	5.000000	0.200000	29.364000
max	99301.000000	22638.480000	14.000000	0.800000	8399.976000

In [7]:
```
df.isnull().sum()
```

Out[7]:
```
Ship Mode       0
Segment         0
Country         0
City            0
State           0
Postal Code     0
Region          0
Category        0
Sub-Category    0
Sales           0
Quantity        0
Discount        0
Profit          0
dtype: int64
```

3. Data Cleaning

In [8]:
```
df.duplicated().sum()
```

Out[8]:
```
np.int64(17)
```

Remove Duplicates

In [9]:
```
df.drop_duplicates(inplace=True)
```

Validate Negative Profit Transactions

In [10]:
```
df[df['Profit'] < 0].head()
```

Out[10]:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Categ
3	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furni
14	Standard Class	Home Office	United States	Fort Worth	Texas	76106	Central	Of Supp
15	Standard Class	Home Office	United States	Fort Worth	Texas	76106	Central	Of Supp
23	Second Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Furni
27	Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Furni

Cleaning Summary

In [11]:
```
cleaning_summary = pd.DataFrame({
    "Step": [
```

```
        "Removed duplicates",
        "Checked missing values",
        "Validated negative profits"
    ],
    "Status": [
        "Completed",
        "No significant missing values",
        "Reviewed"
    ]
})

cleaning_summary
```

Out[11]:

	Step	Status
0	Removed duplicates	Completed
1	Checked missing values	No significant missing values
2	Validated negative profits	Reviewed

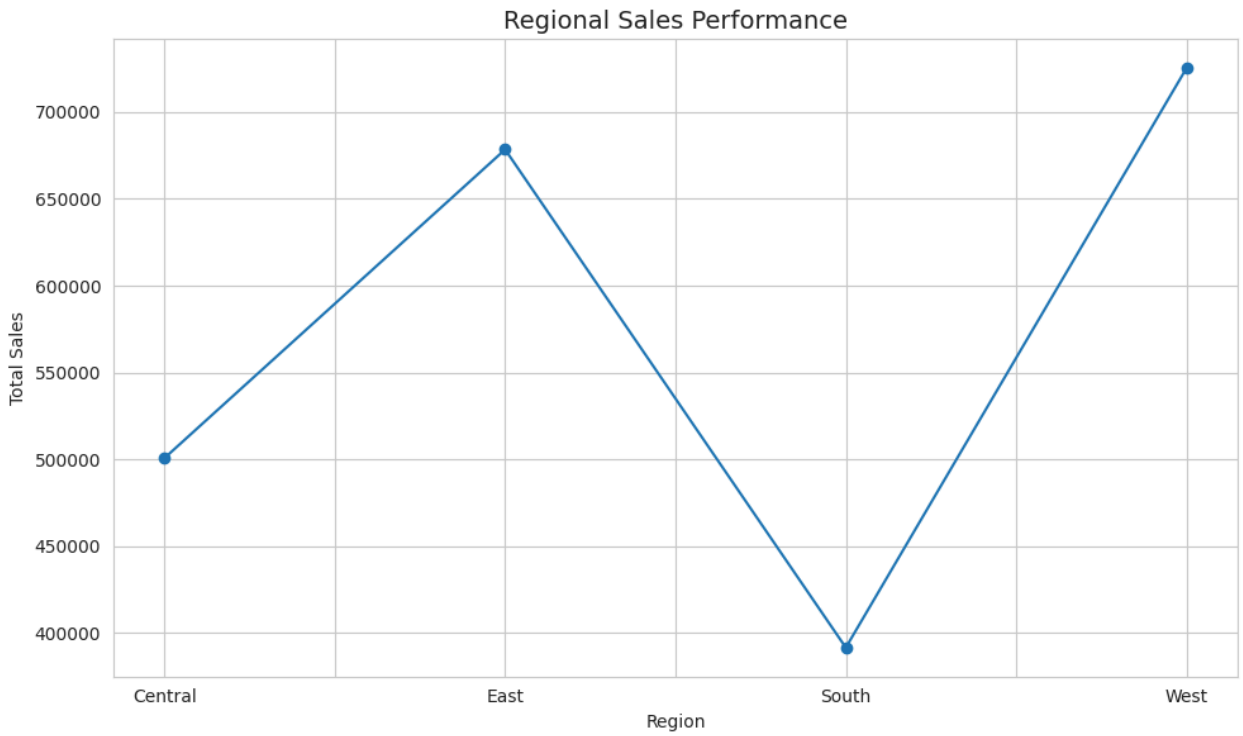
4. Exploratory Data Analysis

📊 6.1 Sales Trend Over Time

In [12]:

```
sales_trend = df.groupby('Region')['Sales'].sum()

plt.figure(figsize=(10,6))
sales_trend.plot(marker='o')
plt.title("Regional Sales Performance", fontsize=14)
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.tight_layout()
plt.show()
```



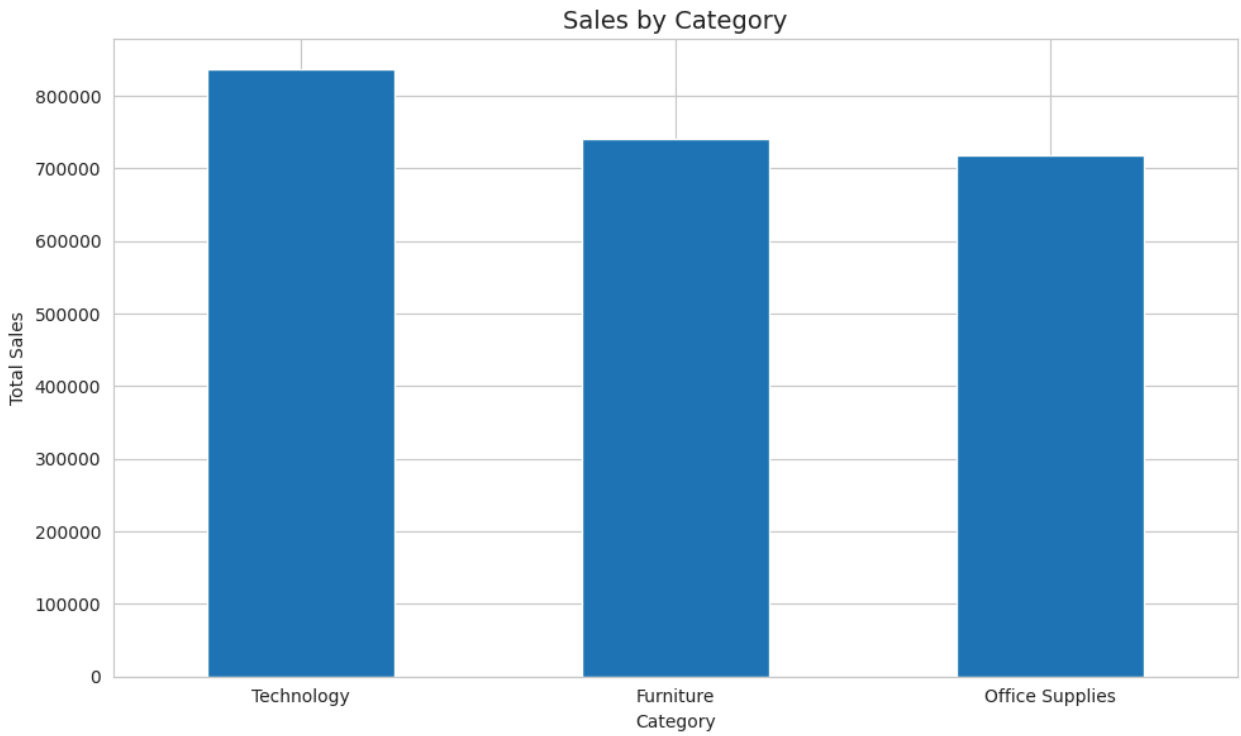
Insight

Sales demonstrate consistent growth East and West regions, indicating positive revenue momentum and market penetration.

📊 6.2 Sales by Category

In [13]:
```
category_sales = df.groupby('Category')['Sales'].sum().sort_values(ascending=F

plt.figure(figsize=(10,6))
category_sales.plot(kind='bar')
plt.title("Sales by Category", fontsize=14)
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



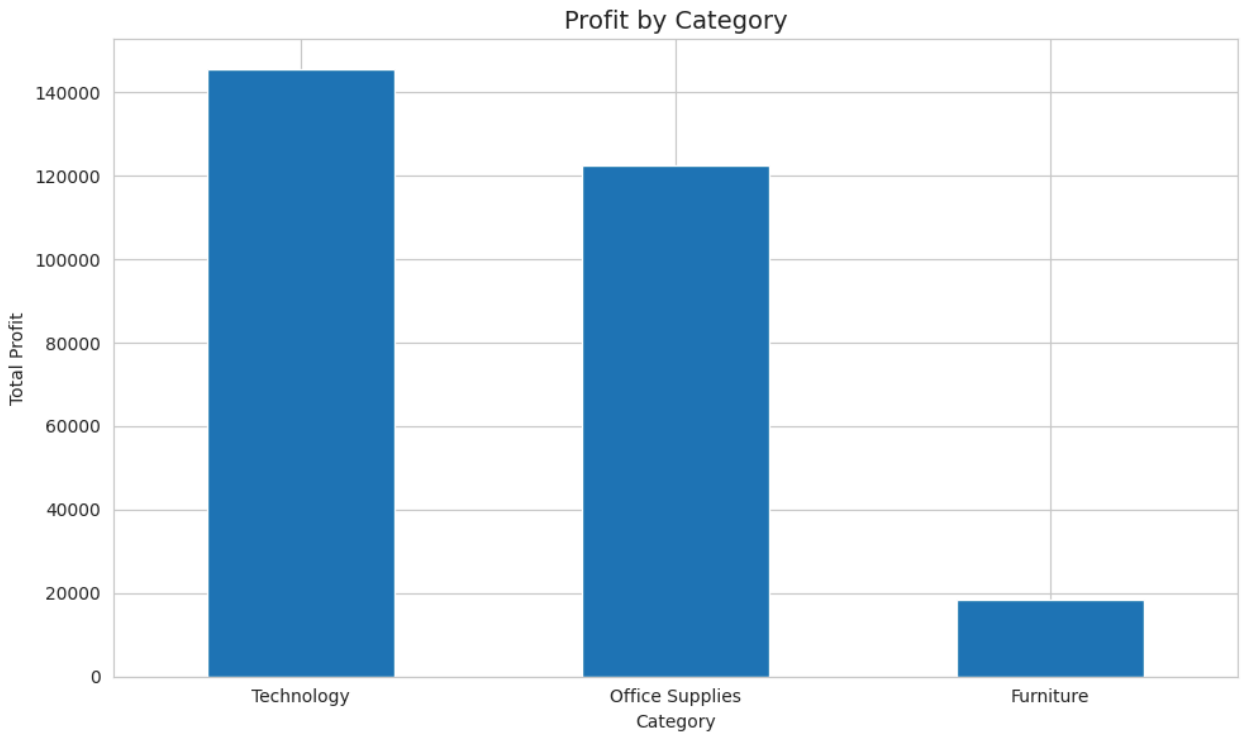
Insight

Technology and Furniture dominate revenue contribution, with Technology generating stronger profitability.

📊 6.3 Profit by Category

In [14]:
```
category_profit = df.groupby('Category')['Profit'].sum().sort_values(ascending

plt.figure(figsize=(10,6))
category_profit.plot(kind='bar')
plt.title("Profit by Category", fontsize=14)
plt.ylabel("Total Profit")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

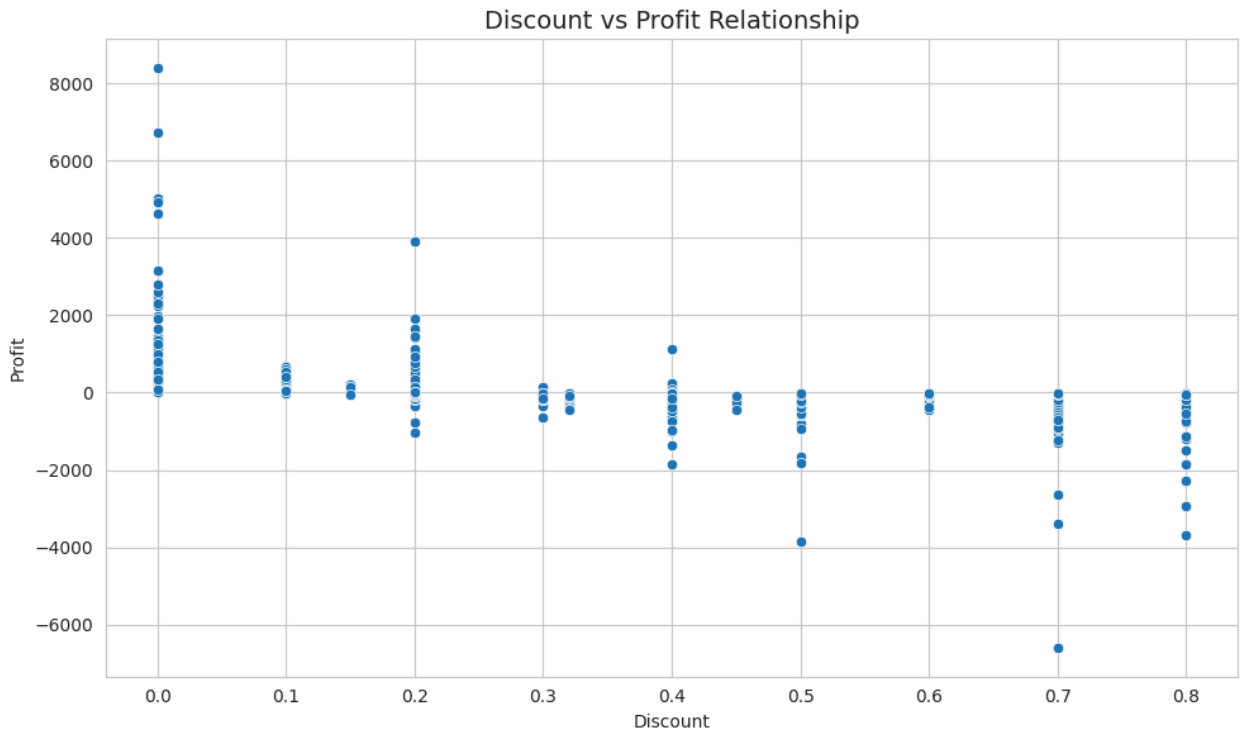


Insight

Technology leads in profit generation, indicating higher margins compared to other categories.

6.4 Discount vs Profit Analysis

In [15]:
```
plt.figure(figsize=(10,6))
sns.scatterplot(x=df['Discount'], y=df['Profit'])
plt.title("Discount vs Profit Relationship", fontsize=14)
plt.tight_layout()
plt.show()
```



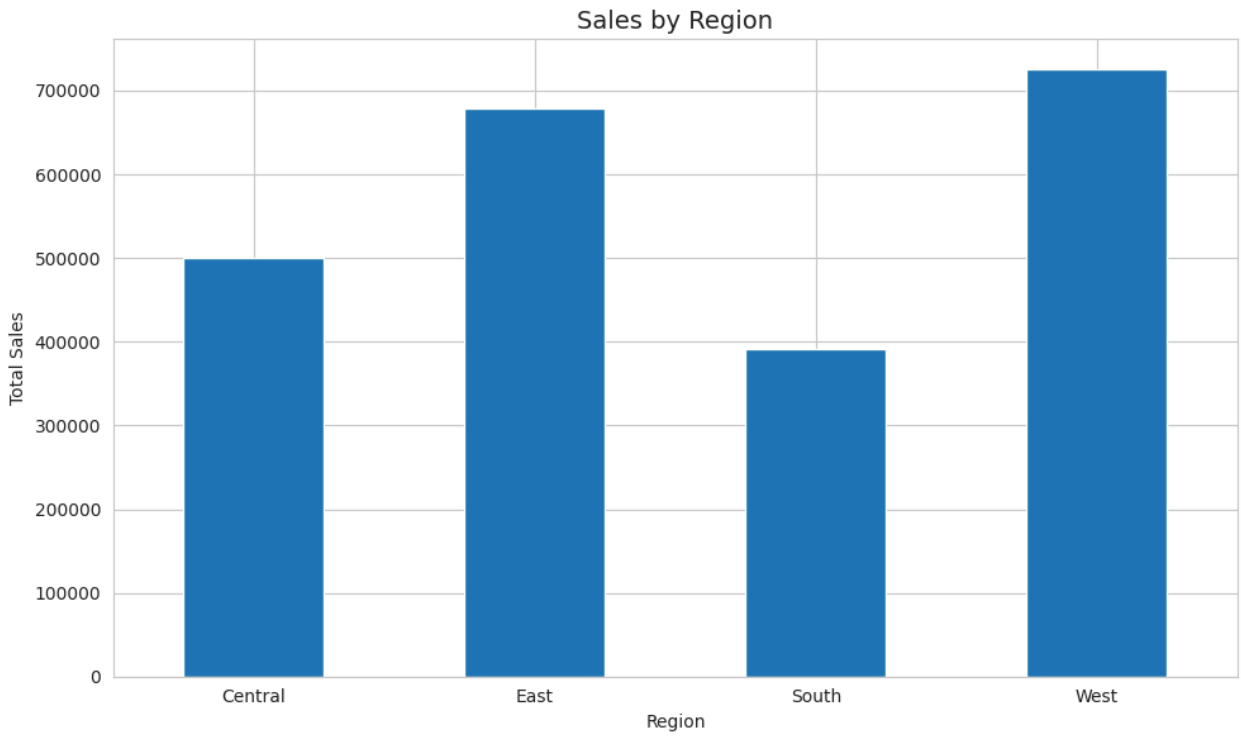
Insight

Higher discount levels correlate with reduced profitability. Aggressive discounting strategies may drive sales volume but negatively impact margins.

6.5 Regional Sales Performance

In [16]:
```
region_sales = df.groupby('Region')['Sales'].sum()

plt.figure(figsize=(10,6))
region_sales.plot(kind='bar')
plt.title("Sales by Region", fontsize=14)
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

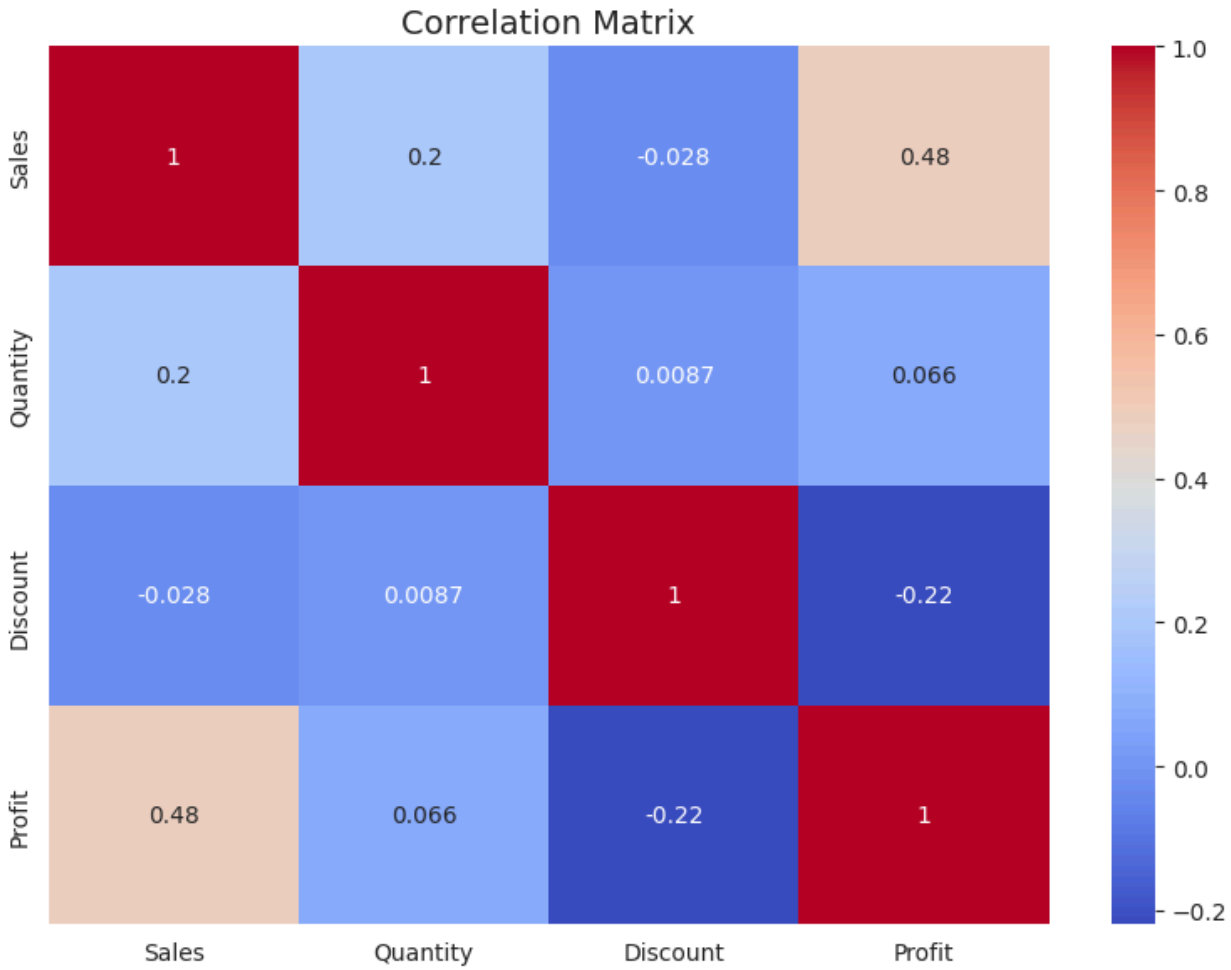


Insight

Regional performance varies significantly, indicating opportunities for region-specific marketing and pricing strategies.

📊 6.6 Correlation Analysis

In [17]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df[['Sales','Quantity','Discount','Profit']].corr(),
            annot=True,
            cmap='coolwarm')
plt.title("Correlation Matrix", fontsize=14)
plt.tight_layout()
plt.show()
```



Insight

Profit is negatively influenced by discount levels, while sales and quantity show positive association.

5. Strategic Insights

1. Revenue growth remains strong, but margin control requires attention.
2. Technology category should be prioritized for expansion.
3. Discount optimization is critical to protect profitability.
4. Revenue concentration suggests focused inventory and marketing strategies.
5. Negative-margin transactions must be monitored and controlled.

6. Conclusion

This analysis demonstrates a structured data analytics workflow from cleaning to

insight generation.

The findings emphasize the importance of balancing revenue growth with margin optimization.

Future enhancements may include: • Customer segmentation analysis • Predictive sales forecasting • Profit optimization modeling • Regional demand forecasting

This project reflects practical business analytics application using real-world retail data.